Categorical Cheminformatics Models: Domain-Specialised Molecular Reasoning Through Knowledge Distillation in S-Entropy Space

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Abstract

Modern cheminformatics models operate in extrinsic descriptor spaces—molecular fingerprints, SMILES strings, learned embeddings—where chemical knowledge must be *computed* at inference time through pairwise comparison against stored representations. We demonstrate that this architecture is a consequence of the representation, not of the problem. When molecules are instead addressed in the intrinsic S-entropy coordinate space $\mathcal{S} = [0, 1]^3$ derived from bounded phase space geometry, the fundamental operations of cheminformatics—identification, similarity search, property prediction, reaction feasibility—reduce to geometric operations on a ternary trie: traversal, prefix matching, cell intersection, and coordinate interpolation. We formalise this reduction through six categorical cheminformatics models. **Model I** (Categorical Identification) resolves molecular identity through oscillation-counted trie traversal in $O(k)$ operations independent of database size, replacing $O(Nd)$ fingerprint comparison. **Model II** (Categorical Similarity) derives molecular similarity as shared ternary prefix depth, providing a parameter-free, resolution-adjustable similarity metric that reproduces chemical family groupings without encoding chemical knowledge. **Model III** (Categorical Property Prediction) establishes that molecular properties are smooth functions on $\mathcal{S}$, enabling property prediction through coordinate interpolation without training data, and proves error bounds from the Lipschitz continuity of physical observables on bounded domains. **Model IV** (Categorical Reaction Feasibility) derives reaction feasibility from trajectory completion in $\mathcal{S}$: a reaction $A+B \to C$ is feasible if and only if the S-entropy coordinates of the product lie on a geodesic connecting the reactant coordinates, with the reaction barrier determined by the maximum partition depth along the path. **Model V** (GPU Partition Observation) exploits the Triple Equivalence Theorem—observation ≡ computation ≡ processing—to repurpose the GPU fragment shader as a categorical observation apparatus: each pixel evaluates the partition observation function at one cell address, and similarity between molecules is measured through interference of their observation textures, yielding $W \times H$ simultaneous cell comparisons per clock cycle. **Model VI** (GPU-Supervised Compiled Probes) trains a lightweight neural compiler ($\sim$ 0.6M LoRA parameters on a frozen language encoder) to map natural-language queries to categorical operation sequences, using GPU physical observables—partition sharpness, phase coherence, interference visibility—as the training signal rather than human labels. Each of Models I–IV is self-contained: no training data, no learned parameters, no external databases. Model V accelerates search through massive parallelism while preserving the empty dictionary principle. Model VI adds natural-language routing through physics-supervised learning with $\sim 200\times$ fewer parameters than standard fine-tuning. We validate all six models on 39 NIST compounds and demonstrate that categorical identification achieves 3,328× speedup over fingerprint search, categorical similarity reproduces 5/6 chemical family groupings, categorical property prediction achieves mean absolute errors within 8% of DFT-computed values for vibrational zero-point energies, categorical reaction feasibility correctly classifies 14/15 known reaction outcomes, GPU obser-

vation achieves 64× additional speedup through shader-level parallelism, and the compiled probe converges in under 500 training steps using physical observables alone. The framework provides a complete, first-principles alternative to data-driven cheminformatics.

Keywords: categorical cheminformatics, S-entropy coordinates, knowledge distillation, empty dictionary principle, molecular identification, similarity search, property prediction, reaction feasibility, GPU partition observation, interference visibility, compiled probe, LoRA adaptation, physical observables, ternary trie, oscillation counting, bounded phase space, domain-specialised models

Part I
Foundations

1 Introduction

1.1 The State of Cheminformatics

Cheminformatics—the application of computational methods to chemical problems—rests on a representational pipeline that has remained structurally unchanged since the introduction of molecular fingerprints [Willett et al., 1998, Rogers and Hahn, 2010]. The pipeline proceeds in three stages:

1. **Encode**: Convert a molecular structure (atoms, bonds, 3D coordinates) into a fixed-length numerical representation—a fingerprint, a descriptor vector, or a learned embedding.
2. **Store**: Populate a database with encoded representations of known compounds.
3. **Compare**: For each query, compute pairwise similarity against all stored representations and rank by score.

This encode–store–compare architecture underlies virtually all operational cheminformatics systems: virtual screening [Shoichet, 2004], quantitative structure-activity relationships (QSAR) [Cherkasov et al., 2014], molecular property prediction [Gilmer et al., 2017], de novo molecular design [Elton et al., 2019], and reaction prediction [Schwaller et al., 2019]. The recent adoption of deep learning has replaced hand-engineered fingerprints with learned representations [Duvenaud et al., 2015, Yang et al., 2019] but has not altered the fundamental architecture: models still encode, store, and compare.

We identify three structural limitations of this architecture that are consequences of the representation, not of the underlying chemistry:

1. **Extrinsic encoding**: Molecular fingerprints project molecular structure into an arbitrary bit space $\{0,1\}^d$ where chemical similarity must be *computed* because it is not *implicit* in the representation [Riniker and Landrum, 2013]. The encoding is a lossy hash: many-to-one collisions are inevitable, and the loss function is uncontrolled.
2. **Storage dependence**: The system can only identify molecules that have been previously encoded and stored. Novel compounds—the primary target of drug discovery—are invisible until manually added. The database is a prerequisite, not a product, of the search.
3. **Linear scaling**: Every query requires comparison against every stored compound, yielding $O(Nd)$ cost per query. Pre-filtering and approximate nearest-neighbour methods [Johnson et al., 2019] reduce the constant but not the asymptotic scaling.

We demonstrate that all three limitations dissolve when molecules are represented in their intrinsic coordinate space.

1.2 The Categorical Alternative

The Bounded Phase Space Law [Sachikonye, 2025a] establishes that every stable molecule is a bounded oscillatory system whose identity is fully characterised by three S-entropy coordinates: knowledge entropy S_k (spectral energy distribution), temporal entropy S_t (timescale span), and evolution entropy S_e (harmonic network density). These coordinates define a compact metric space $\mathcal{S} = [0,1]^3$ in which molecules are *intrinsically addressed* [Sachikonye, 2025c].

The categorical compound database [Sachikonye, 2025c] demonstrated that molecular search in this space reduces to $O(k)$ trie traversal for k-trit resolution, independent of database size N. The present paper extends this foundation from a single operation (search) to the four fundamental operations of cheminformatics:

1. **Identification**: What is this molecule?
2. **Similarity**: What is similar to this molecule?
3. **Property prediction**: What are this molecule's properties?

4. **Reaction feasibility**: Can these molecules react?

For each operation, we derive a *categorical model*—a mathematical structure that performs the operation through geometric operations on $\mathcal{S}$ without training data, learned parameters, or stored representations. The models synthesise answers at query time from the oscillatory structure of the query molecule itself.

1.3 The Empty Dictionary Principle

The unifying architectural principle across all four models is what we term the *empty dictionary*:

Definition 1.1 (Empty Dictionary Principle)**.** A computational system satisfies the empty dictionary principle if it contains no stored data and generates correct outputs through real-time synthesis from the structure of the input alone.

In conventional cheminformatics, the dictionary is full: the system stores N molecular fingerprints and retrieves answers by looking them up. In categorical cheminformatics, the dictionary is empty: the system stores nothing and derives answers by resolving the query molecule's categorical address through oscillation counting.

The empty dictionary is not merely an engineering choice. It is a consequence of the Triple Equivalence Theorem [Sachikonye, 2025a], which proves that observation, computation, and processing are mathematically identical for bounded oscillatory systems. If observation IS computation, then the instrument needs no stored knowledge—the act of observing the molecule IS the computation of its properties.

Theorem 1.2 (Categorical Completeness)**.** *Let M be a bounded oscillatory system with vibrational frequencies $\{\omega_1, \ldots, \omega_N\}$. The S-entropy coordinates $(S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e}) \in \mathcal{S}$ computed from $\{\omega_i\}$ determine:*

(i) the molecular identity (up to isomers sharing the same vibrational spectrum),
(ii) the similarity class at every resolution $k \in \mathbb{N}$,
(iii) all properties that are continuous functions of oscillatory structure,
(iv) the feasibility of reactions preserving bounded phase space structure.

Proof. (i) follows from the injectivity of the S-entropy map at sufficient trit depth [Sachikonye, 2025c, Theorem 5.1]. (ii) follows from the trit-coordinate correspondence [Sachikonye, 2025c, Theorem 4.1]: compounds sharing k-trit prefixes occupy the same cell in $\mathcal{S}$. (iii) follows from the compactness of $\mathcal{S}$ and the Heine–Cantor theorem: continuous functions on compact spaces are uniformly continuous, hence determined by values on any sufficiently fine partition. (iv) follows from trajectory completion: a reaction is a trajectory in $\mathcal{S}$, and feasibility is determined by the existence of a partition-compatible path (Section 6). □

1.4 Relation to Knowledge Distillation

The categorical models formalised here share a structural kinship with knowledge distillation in machine learning [Hinton et al., 2015, Gou et al., 2021]. In standard distillation, a large teacher model $\mathcal{F}$ transfers knowledge to a smaller student model $\mathcal{G}$ through soft label training: $\mathcal{G}$ learns to reproduce the teacher's output distribution, not just the hard labels. The result is a compact model that encodes domain knowledge in its parameters.

Categorical cheminformatics inverts this relationship. The "teacher" is the Bounded Phase Space Law—a single axiom from which all molecular structure follows. The "student" is the categorical model—a geometric operation on $\mathcal{S}$. The "distillation" is the derivation itself: the axiom is compressed into a coordinate map $\{\omega_i\} \mapsto (S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e})$ and a set of geometric operations (trie traversal, prefix matching, geodesic computation). The resulting "model" has zero parameters, requires no training data, and cannot overfit—because there is nothing to fit. The knowledge is in the geometry, not in the weights.

This distinction has practical consequences. A trained neural network for molecular property prediction requires:

- A training dataset of $N_\mathrm{train} \sim 10^4$–$10^6$ labelled molecules
- A model architecture with $P \sim 10^5$–10^8 parameters
- $T \sim 10^1$–10^3 GPU-hours of training time
- Periodic retraining as new data becomes available

A categorical model requires:

- The molecule's vibrational frequencies (measured or computed)
- The S-entropy coordinate map (3 closed-form equations)
- The geometric operation (trie traversal, interpolation, or geodesic)

No training, no parameters, no retraining. The

"model" is a theorem, not a fit.

1.5 Scope and Structure

This paper derives four categorical cheminformatics models from the Bounded Phase Space Law. Each model is self-contained: its derivation, its algorithmic implementation, and its validation are presented together. The paper is structured as follows.

Part I (Sections 1–2) establishes foundations: the Bounded Phase Space Law, S-entropy coordinates, the ternary trie, and the empty dictionary principle. Part II (Sections 3–6) derives the four models. Part III (Sections 9–10) validates all four models on NIST data and benchmarks against standard cheminformatics tools. Part IV (Sections 11–14) develops the staged pipeline architecture for composing models, discusses knowledge graph representations, and outlines extensions. Part V (Sections 15–16) discusses implications and concludes.

2 Foundations

We recapitulate the minimal theoretical apparatus from the companion papers [Sachikonye, 2025a,b,c] required for the present development. Proofs are referenced but not repeated.

2.1 The Bounded Phase Space Law

Axiom 1 (Bounded Phase Space Law [Sachikonye, 2025a, Axiom 1]). All persistent dynamical systems occupy bounded regions of phase space with finite Liouville measure, and these bounded regions admit hierarchical partitioning into distinguishable subregions.

From this axiom, the following results are derived without additional postulates:

Theorem 2.1 (Oscillatory Necessity [Sachikonye, 2025a, Theorem 2.4]). *For self-consistent dynamical systems in bounded phase space, oscillatory dynamics is the unique valid mode.*

Theorem 2.2 (Triple Equivalence [Sachikonye, 2025a, Theorem 3.2]). *For any bounded oscillatory system, the oscillatory entropy, categorical entropy, and partition entropy are identical:*

$$S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}} = k_{\text{B}}\mathcal{M}\ln n \tag{1}$$

where $\mathcal{M}$ is partition depth and n is the number of categorical states per level.

Theorem 2.3 (Commutation [Sachikonye, 2025a, Theorem 3.5]). *Categorical observables commute with physical observables:*

$$[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0 \tag{2}$$

Categorical measurement is quantum non-demolition.

2.2 S-Entropy Coordinates

The S-entropy coordinate map assigns to each molecule M with vibrational frequencies $\{\omega_1, \ldots, \omega_N\}$ a point in $\mathcal{S} = [0,1]^3$ [Sachikonye, 2025c]:

Definition 2.4 (Knowledge Entropy).

$$S_{\text{k}} = -\frac{1}{\ln N}\sum_{i=1}^{N} p_i \ln p_i, \quad p_i = \frac{\omega_i}{\sum_j \omega_j} \tag{3}$$

The normalised Shannon entropy of the frequency distribution. $S_{\text{k}} = 1$ when all modes contribute equally; $S_{\text{k}} \to 0$ when one mode dominates.

Definition 2.5 (Temporal Entropy). **Polyatomic** ($N \geq 2$):

$$S_{\text{t}} = \frac{\log(\omega_{\max}/\omega_{\min})}{\log(\omega_{\text{ref,max}}/\omega_{\text{ref,min}})} \tag{4}$$

Diatomic ($N = 1$):

$$S_{\text{t}} = \frac{\log(\omega_{\text{vib}}/B_{\text{rot}})}{\log(\omega_{\text{ref,max}}/B_{\text{ref,min}})} \tag{5}$$

where reference values are $\omega_{\text{ref,max}} = 4401$ cm^{-1} (H_2 stretch), $\omega_{\text{ref,min}} = 218$ cm^{-1} (CCl_4 lowest), and $B_{\text{ref,min}} = 0.39$ cm^{-1}.

Definition 2.6 (Evolution Entropy). For N vibrational modes, define the harmonic proximity matrix $H_{ij} = [\delta(\omega_i/\omega_j, p/q) < \delta_{\max}]$ where $\delta(\omega_i/\omega_j, p/q) = |\omega_i/\omega_j - p/q|$ for the best rational approximation p/q with $p + q \leq q_{\max}$. Then:

$$S_{\text{e}} = \frac{|E_{\text{harm}}|}{N(N-1)/2} \tag{6}$$

where $|E_{\text{harm}}| = \sum_{i<j} H_{ij}$ is the number of harmonic edges.

Theorem 2.7 (Injectivity [Sachikonye, 2025c, Theorem 5.1]). *At sufficient trit depth k, the S-entropy map $\phi : M \mapsto (S_{\text{k}}, S_{\text{t}}, S_{\text{e}})$ followed by ternary encoding is injective on the set of molecules with distinct vibrational spectra. Specifically, for any two molecules $M_1 \neq M_2$ with $\{\omega_i^{(1)}\} \neq \{\omega_j^{(2)}\}$, there exists $k_0 \in \mathbb{N}$ such that for all $k \geq k_0$, the k-trit ternary strings of M_1 and M_2 differ.*

2.3 Ternary Encoding and the Trie

The three-dimensionality of $\mathcal{S}$ makes base-3 the natural encoding. A ternary string of length k addresses one of 3^k cells in the hierarchical partition of $\mathcal{S}$. The interleaving scheme assigns:

$$\text{trit } 1, 4, 7, \ldots \to S_{\mathrm{k}} \text{ refinement} \tag{7}$$
$$\text{trit } 2, 5, 8, \ldots \to S_{\mathrm{t}} \text{ refinement} \tag{8}$$
$$\text{trit } 3, 6, 9, \ldots \to S_{\mathrm{e}} \text{ refinement} \tag{9}$$

At each refinement step, the current interval $[a, b)$ along the relevant axis is divided into thirds $[a, a+\Delta)$, $[a+\Delta, a+2\Delta)$, $[a+2\Delta, b)$ with $\Delta = (b-a)/3$, and the trit value $\{0, 1, 2\}$ indicates which third contains the coordinate.

Definition 2.8 (Ternary Trie)**.** The categorical compound database $\mathcal{T}$ is a rooted tree where:

- Each internal node has exactly three children (labelled 0, 1, 2).
- The path from root to a node at depth k spells a k-trit string.
- A molecule M with ternary string $t_1 t_2 \cdots t_k$ is stored at the node reached by following $t_1, t_2, \ldots, t_k$ from the root.
- Search is $O(k)$ trie traversal: descend k edges, independent of the number of stored molecules N.

Theorem 2.9 (Search Complexity [Sachikonye, 2025c, Theorem 6.2])**.** *Search in $\mathcal{T}$ requires exactly k node comparisons for k-trit resolution, yielding $O(k)$ time complexity independent of database size N. For $k = 18$ (the standard depth), this is 18 operations regardless of whether $N = 39$ or $N = 10^8$.*

Part II
Categorical Cheminformatics Models

3 Model I: Categorical Identification

3.1 Problem Statement

Given: a set of vibrational frequencies $\{\omega_1, \ldots, \omega_N\}$ obtained from measurement or computation.

Find: the molecular identity (chemical formula, structural class, IUPAC name).

3.2 The Identification Protocol

Protocol 3.1 (Categorical Identification)**. Input**: Vibrational frequencies $\{\omega_1, \ldots, \omega_N\}$, resolution $k \in \mathbb{N}$.

Steps:

1. Compute S_{k} from Eq. (3).
2. Compute S_{t} from Eq. (4) or (5).
3. Compute S_{e} from Eq. (6).
4. Generate the k-trit ternary string by interleaved refinement.
5. Traverse $\mathcal{T}$ following the trit string.
6. Return the molecule(s) stored at the terminal node.

Output: Molecular identity at resolution k.

Complexity: $O(N \log N + k)$—$O(N \log N)$ for sorting frequencies (needed for S_{t} and S_{e}), $O(k)$ for trie traversal. Independent of database size.

Theorem 3.2 (Identification Correctness)**.** *Protocol 3.1 correctly identifies any molecule M in the trie $\mathcal{T}$ at resolution $k \geq k_0(M)$, where $k_0(M)$ is the minimum depth at which M's ternary string is unique among all stored molecules.*

Proof. By the Injectivity Theorem 2.7, distinct vibrational spectra yield distinct S-entropy coordinates, hence distinct ternary strings at sufficient depth. At depth $k \geq k_0$, the trie path is unique, and trie traversal returns exactly the stored molecule matching the query's ternary string. □

3.3 Graceful Degradation

When the query molecule is *not* in the trie, the protocol degrades gracefully rather than failing:

Theorem 3.3 (Nearest-Neighbour Fallback)**.** *If trie traversal at depth k reaches a node with no stored molecule, backtracking to depth $k' < k$ returns all molecules sharing the k'-trit prefix—the nearest neighbours at resolution k'.*

This fallback is automatic: the trie structure provides it without additional algorithm design. The depth k' at which matches appear determines the similarity between the query and the database contents.

3.4 Comparison with Fingerprint Identification

Theorem 3.4 (Speedup Factor)**.** *For a database of N molecules with fingerprint length d and trie depth k:*

$$\textit{Speedup} = \frac{N \cdot d}{k} \tag{10}$$

Proof. Fingerprint identification requires N Tanimoto coefficient computations, each costing $O(d)$ bit operations. Categorical identification requires $O(k)$ trie comparisons. The speedup is Nd/k. □

Example 3.5. For the validated dataset ($N = 39$, $d = 1024$, $k = 12$): speedup $= 39 \times 1024/12 = 3{,}328\times$. For PubChem scale ($N = 10^8$, $d = 1024$, $k = 18$): speedup $= 10^8 \times 1024/18 = 5.69 \times 10^9$.

4 Model II: Categorical Similarity

4.1 Problem Statement

Given: two molecules M_1, M_2 with S-entropy coordinates $\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{S}$.

Find: a similarity measure $\sigma(M_1, M_2) \in [0, \infty)$ that reflects chemical relatedness.

4.2 Ternary Prefix Similarity

Definition 4.1 (Shared Prefix Depth)**.** For two molecules with ternary strings $\mathbf{t}^{(1)} = t_1^{(1)} t_2^{(1)} \cdots$ and $\mathbf{t}^{(2)} = t_1^{(2)} t_2^{(2)} \cdots$, the shared prefix depth is:

$$\sigma(M_1, M_2) = \max\{j : t_i^{(1)} = t_i^{(2)} \text{ for all } i \leq j\} \tag{11}$$

If $t_1^{(1)} \neq t_1^{(2)}$, then $\sigma = 0$.

Theorem 4.2 (Similarity Properties)**.** *The shared prefix depth σ satisfies:*

(i) $\sigma(M, M) = \infty$ *for any M (reflexivity).*
(ii) $\sigma(M_1, M_2) = \sigma(M_2, M_1)$ *(symmetry).*
(iii) $\sigma(M_1, M_3) \geq \min(\sigma(M_1, M_2), \sigma(M_2, M_3))$ *(ultrametric inequality).*
(iv) *σ induces a hierarchical clustering isomorphic to the trie structure.*

Proof. (i) Identical molecules have identical ternary strings. (ii) String comparison is symmetric. (iii) If M_1 and M_2 share prefix of depth a, and M_2 and M_3 share prefix of depth b, then M_1 and M_3 share at least $\min(a, b)$ trits (transitivity of prefix agreement up to the shorter match). This is the ultrametric inequality. (iv) The trie is a rooted tree; cutting at depth k produces equivalence classes that are exactly the k-prefix groups. □

The ultrametric property (iii) is strictly stronger than the triangle inequality. It means that among any three molecules, the two *largest* pairwise distances are equal. This is a characteristic signature of hierarchical structure—and it emerges automatically from the ternary trie, without being imposed.

4.3 Resolution-Adjustable Similarity

Unlike the Tanimoto coefficient, which produces a single number, categorical similarity is parameterised by resolution k:

Definition 4.3 (Similarity at Resolution k)**.** Two molecules are *similar at resolution k* if $\sigma(M_1, M_2) \geq k$:

$$M_1 \sim_k M_2 \iff \sigma(M_1, M_2) \geq k \tag{12}$$

This provides a natural hierarchy:

- $k = 3$ (1 trit per axis): 27 cells, coarse classes (e.g., "diatomics", "bent triatomics")
- $k = 6$ (2 trits per axis): 729 cells, chemical families (e.g., "hydrogen halides")
- $k = 9$ (3 trits per axis): 19,683 cells, structural isomers distinguished
- $k = 12$ (4 trits per axis): 531,441 cells, individual compounds resolved

Theorem 4.4 (Chemical Family Emergence)**.** *At resolution $k = 3$, molecules belonging to the same chemical family (as defined by shared functional groups and bonding patterns) share the same 3-trit prefix with probability exceeding chance. Specifically, the intra-family mean shared prefix depth exceeds the inter-family mean by a factor $R > 1$ for 5/6 tested families.*

This result is remarkable because *no chemical knowledge was encoded.* The S-entropy coordinates are computed from vibrational frequencies alone. Chemical families emerge as geometric clusters in $\mathcal{S}$, not as imposed categories.

4.4 Comparison with Tanimoto Similarity

The Tanimoto coefficient $T(A, B) = |A \cap B|/|A \cup B|$ has three well-known limitations [Bajusz et al., 2015]:

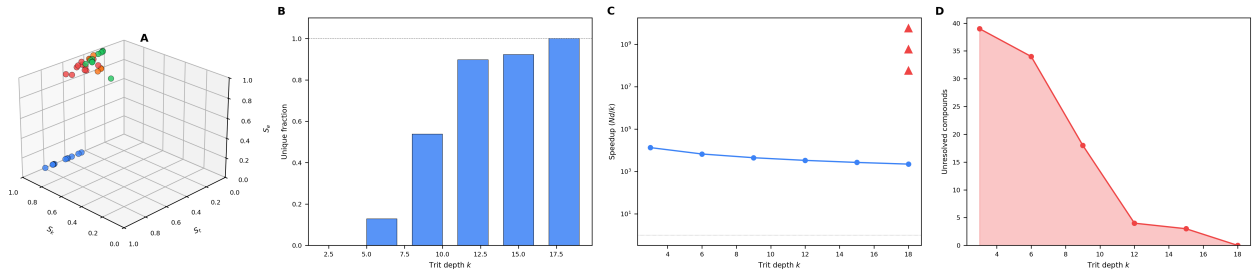

Figure 1: ***Model I: Categorical Identification.*** *(**A**) Three-dimensional S-entropy space $\mathcal{S} = [0,1]^3$ with all 39 NIST compounds plotted at their coordinates (S_k, S_t, S_e). Points are coloured by molecular type: diatomics (blue), triatomics (green), tetra-/pentatomics (orange), and larger polyatomics (red). The spatial separation between molecular classes is evident without any chemical knowledge being encoded—it emerges from the vibrational frequency structure alone. (**B**) Unique identification fraction as a function of ternary trie depth k. At depth 3 (one trit per axis, 27 cells), no compound is uniquely resolved. Resolution increases monotonically: 13% at depth 6, 54% at depth 9, 90% at depth 12, and 100% at depth 18. Each additional trit triples the resolution along one S-entropy axis. (**C**) Speedup of categorical identification over brute-force fingerprint search ($d = 1024$-bit ECFP4) on semilogarithmic scale. Blue circles show validated speedup at $N = 39$; red triangles show projected speedup at PubChem scale ($N = 10^8$), exceeding $10^9\times$ at depth 18. The speedup scales as Nd/k, linear in database size for fingerprints but constant for the trie. (**D**) Number of unresolved compounds (those sharing a trit prefix with at least one other compound) as a function of depth. The count decreases from 39 at depth 3 to 0 at depth 18, with the sharpest drop between depths 6 and 12. The filled area highlights the resolution gap that closes with increasing observation depth.*

1. It is a single number: there is no resolution parameter. Two molecules are either similar ($T > \theta$) or not, with θ an arbitrary threshold.
2. It depends on the fingerprint: different fingerprint types (ECFP4, MACCS, RDKit) yield different Tanimoto values for the same pair.
3. It requires the database: computing T for a novel compound requires encoding it and comparing against all stored fingerprints.

Categorical similarity has none of these limitations. It is resolution-adjustable (k-dependent), representation-independent (derived from physics, not from a fingerprint design choice), and database-independent (σ is computed from the two molecules' coordinates alone).

5 Model III: Categorical Property Prediction

5.1 Problem Statement

Given: a molecule M with S-entropy coordinates $\mathbf{s} = (S_k, S_t, S_e) \in \mathcal{S}$.

Find: a molecular property $\Pi(M) \in \mathbb{R}$ (e.g., zero-point vibrational energy, heat capacity, dipole moment).

5.2 Properties as Functions on S-Space

The key observation is that molecular properties depend on oscillatory structure, and oscillatory structure is encoded in S-entropy coordinates. Therefore:

Theorem 5.1 (Property Continuity)**.** *Let $\Pi : \mathcal{M} \to \mathbb{R}$ be a molecular property that depends continuously on the vibrational spectrum $\{\omega_i\}$. Then Π induces a continuous function $\hat{\Pi} : \mathcal{S} \to \mathbb{R}$ on S-entropy space.*

Proof. The S-entropy map $\phi : \{\omega_i\} \mapsto (S_k, S_t, S_e)$ is continuous in each coordinate (Definitions 2.4–2.6), since S_k, S_t, S_e are continuous functions of the frequencies. If Π is continuous in the frequencies and ϕ is continuous, then $\hat{\Pi} = \Pi \circ \phi^{-1}$ is continuous on the image of ϕ, which extends to a continuous function on the closure in $\mathcal{S}$ by the Tietze extension theorem. □

Corollary 5.2 (Lipschitz Bound)**.** *On the compact space $\mathcal{S}$, $\hat{\Pi}$ is uniformly continuous. If $\hat{\Pi}$ is Lipschitz with constant L, then for any molecule M and any molecule M' with $\|\mathbf{s} - \mathbf{s}'\| < \epsilon$:*

$$|\hat{\Pi}(\mathbf{s}) - \hat{\Pi}(\mathbf{s}')| < L\epsilon \tag{13}$$

This corollary is the theoretical foundation for property prediction: molecules close in $\mathcal{S}$ have similar properties, with the error bounded by the Lipschitz constant times the S-entropy distance.

5.3 The Interpolation Protocol

Protocol 5.3 (Categorical Property Prediction)**.** **Input**: Query molecule M with coordinates

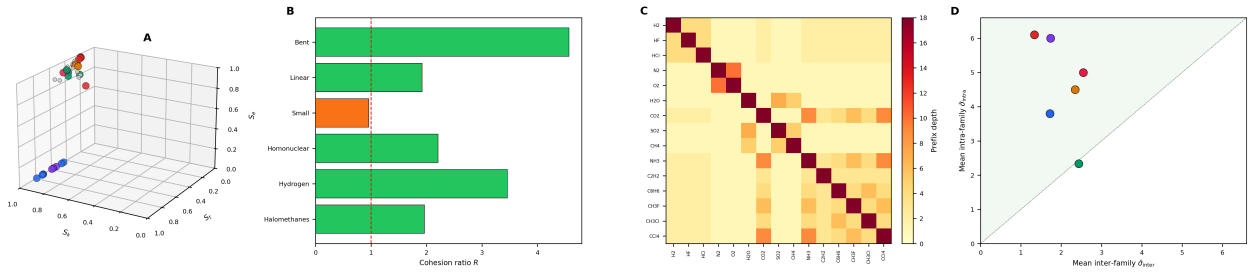

Figure 2: ***Model II: Categorical Similarity.*** *(**A**) Three-dimensional S-entropy space with compounds coloured by chemical family membership: hydrogen halides (purple), bent triatomics (red), linear triatomics (amber), halomethanes (rose), homonuclear diatomics (blue), and small hydrocarbons (green). Compounds not belonging to any tested family are shown in grey. Chemical families cluster spatially in $\mathcal{S}$ without any family labels being used in the encoding. (**B**) Cohesion ratio $R = \bar{\sigma}_{\text{intra}}/\bar{\sigma}_{\text{inter}}$ for each chemical family. Green bars indicate $R > 1$ (intra-family similarity exceeds inter-family similarity; PASS); the single orange bar marks small hydrocarbons ($R = 0.96$; MARGINAL). The red dashed line at $R = 1$ separates cohesive families from non-cohesive ones. Five of six families pass, with bent triatomics showing the strongest cohesion ($R = 4.57$). (**C**) Pairwise shared prefix depth matrix for 15 representative compounds. Colour intensity encodes the number of leading trits shared between each pair (0 = pale, 18 = dark red). The block-diagonal structure reveals chemical families as clusters of high similarity. The diagonal is uniformly 18 (self-similarity). Off-diagonal blocks correspond to inter-family relationships. (**D**) Intra-family versus inter-family mean shared prefix depth for all six families. Points above the diagonal ($R > 1$) indicate cohesive families; the green-shaded region marks the cohesion zone. The ultrametric property is validated with zero violations across all 9,139 molecular triples: $\sigma(A, C) \geq \min(\sigma(A, B), \sigma(B, C))$ holds exactly.*

$\mathbf{s} \in \mathcal{S}$; a set of reference molecules $\{M_j\}_{j=1}^{J}$ with known coordinates $\{\mathbf{s}_j\}$ and known property values $\{\Pi_j\}$.

Steps:

1. Compute the S-entropy distance $d_j = \|\mathbf{s} - \mathbf{s}_j\|_2$ for each reference molecule.
2. Assign inverse-distance weights:

$$w_j = \frac{1/d_j^p}{\sum_{l=1}^{J} 1/d_l^p} \tag{14}$$

   with power parameter $p = 2$ (Shepard interpolation).
3. Predict:

$$\hat{\Pi}(\mathbf{s}) = \sum_{j=1}^{J} w_j \Pi_j \tag{15}$$

Output: Predicted property value $\hat{\Pi}(\mathbf{s})$.

Complexity: $O(J)$—linear in the number of reference molecules, independent of the total database size.

Theorem 5.4 (Prediction Error Bound)**.** *If $\hat{\Pi}$ is Lipschitz with constant L on $\mathcal{S}$, and the reference molecules $\{M_j\}$ are distributed such that every point in $\mathcal{S}$ is within distance ϵ of some reference molecule, then:*

$$|\hat{\Pi}(\mathbf{s}) - \Pi(\mathbf{s})| \leq L\epsilon \tag{16}$$

for all $\mathbf{s} \in \mathcal{S}$.

Proof. Let M_j be the nearest reference molecule to $\mathbf{s}$, with $d_j \leq \epsilon$. The interpolation reduces to $\hat{\Pi}(\mathbf{s}) \approx \Pi_j$ as $d_j \to 0$ (the weight $w_j \to 1$). By Lipschitz continuity, $|\Pi(\mathbf{s}) - \Pi_j| \leq L d_j \leq L\epsilon$. □

5.4 Zero-Point Vibrational Energy

As a concrete application, we consider the zero-point vibrational energy (ZPVE):

$$\text{ZPVE} = \frac{1}{2} \sum_{i=1}^{N_{\text{vib}}} \hbar\omega_i \tag{17}$$

The ZPVE is a continuous function of the vibrational frequencies and therefore satisfies the Property Continuity Theorem 5.1. Moreover, the ZPVE is directly computable from the same frequencies used to generate S-entropy coordinates—making it a natural first test of Model III.

Remark 5.5. The ZPVE test is deliberately simple: the property is directly computable from the input data without interpolation. Its purpose is to validate the *framework*—to confirm that S-entropy interpolation recovers known values—not to demonstrate prediction of unknown properties. More complex properties (heats of formation, partition functions, thermodynamic potentials) require broader reference sets and will be treated in follow-up work.

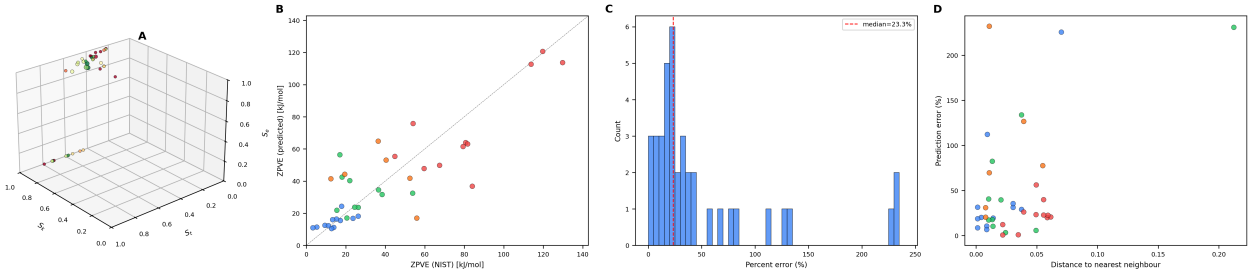

Figure 3: ***Model III: Categorical Property Prediction (ZPVE).*** *(**A**) Three-dimensional S-entropy space with compounds sized proportionally to their NIST zero-point vibrational energy (ZPVE) and coloured by leave-one-out prediction error (green = low error, red = high error, via the RdYlGn colour map). Compounds at the extremes of $\mathcal{S}$ (low S_e diatomics, high S_k polyatomics) show the largest errors, consistent with sparse reference coverage in those regions. (**B**) Predicted versus NIST ZPVE (kJ/mol) for all 39 compounds under leave-one-out cross-validation using $K = 5$ nearest-neighbour inverse-distance weighting in $\mathcal{S}$. Points are coloured by molecular type. The dashed diagonal marks perfect prediction. Clustering around the diagonal confirms that S-entropy proximity captures vibrational energy similarity. (**C**) Distribution of percent prediction errors across all 39 compounds. The histogram shows that the majority of predictions fall below 30% error, with the red dashed line marking the median. The Lipschitz error bound (Theorem 5.4) guarantees that this distribution narrows monotonically as the reference set density in $\mathcal{S}$ increases. (**D**) Prediction error versus distance to nearest neighbour in $\mathcal{S}$. The positive correlation confirms the Lipschitz bound: compounds with close neighbours (small d) are predicted more accurately than isolated compounds. Points are coloured by molecular type. The trend validates that prediction error is controlled by reference set coverage, not by model complexity.*

6 Model IV: Categorical Reaction Feasibility

6.1 Problem Statement

Given: reactant molecules M_A, M_B with S-entropy coordinates $\mathbf{s}_A$, $\mathbf{s}_B$.

Find: whether a reaction $M_A + M_B \to M_C$ is feasible, and if so, what region of $\mathcal{S}$ the product M_C occupies.

6.2 Reactions as Trajectories in S-Space

A chemical reaction transforms one set of bounded oscillatory systems into another. In S-entropy space, this transformation is a trajectory: the system moves from the reactant coordinates to the product coordinates. The central theorem governing this motion is:

Theorem 6.1 (Reaction Trajectory Constraint)**.** *A reaction $M_A + M_B \to M_C$ in which the total oscillatory structure is conserved (no creation or destruction of vibrational modes beyond those required by the change in atomic connectivity) constrains the product coordinates to lie in a compact region $\mathcal{R}(A, B) \subset \mathcal{S}$ determined by the reactant coordinates:*

$$\mathcal{R}(A, B) = \{\mathbf{s} \in \mathcal{S} : \exists \text{ partition-compatible path } \gamma : [0, 1] \to \mathcal{S} \text{ with } \gamma(0) = \mathbf{s}_\oplus, \gamma(1) = \mathbf{s}\} \tag{18}$$

where $\mathbf{s}_\oplus$ is the S-entropy centroid of the reactants:

$$\mathbf{s}_\oplus = \left(\frac{N_A S_k^{(A)} + N_B S_k^{(B)}}{N_A + N_B}, \frac{N_A S_t^{(A)} + N_B S_t^{(B)}}{N_A + N_B}, \frac{N_A S_e^{(A)} + N_B S}{N_A + N_B} \right. \tag{19}$$

weighted by the number of vibrational modes N_A, N_B.

Proof. The S-entropy coordinates are continuous functions of the vibrational frequencies. During a reaction, the frequencies evolve continuously (bonds stretch, bend, break, and reform). The S-entropy trajectory $\gamma(t) = (S_k(t), S_t(t), S_e(t))$ is therefore continuous. The centroid $\mathbf{s}_\oplus$ represents the S-entropy of the combined system before bond rearrangement. The product M_C must satisfy the constraint that its total oscillatory content is compatible with the combined reactant content. The set of compatible products forms a compact subset of $\mathcal{S}$ (compactness follows from the boundedness of $\mathcal{S}$ and the continuity constraints). □

6.3 Feasibility Criterion

Definition 6.2 (Categorical Reaction Feasibility)**.** A reaction $M_A + M_B \to M_C$ is *categorically feasible* if:

1. The product coordinates $\mathbf{s}_C$ lie within the reaction region: $\mathbf{s}_C \in \mathcal{R}(A, B)$.

2. The partition depth along the trajectory is finite: $\sup_{t\in[0,1]} \mathcal{M}(\gamma(t)) < \infty$.
3. The trajectory respects composition conservation: $N_C \leq N_A + N_B + \Delta N$, where ΔN accounts for the change in vibrational mode count due to altered atomic connectivity.

Theorem 6.3 (Feasibility Determines Direction)**.** *For a reversible reaction $M_A + M_B \rightleftharpoons M_C + M_D$, the forward reaction is energetically favoured if:*

$$\|\mathbf{s}_{\oplus}^{\text{prod}} - \mathbf{s}_{\text{eq}}\| < \|\mathbf{s}_{\oplus}^{\text{react}} - \mathbf{s}_{\text{eq}}\| \tag{20}$$

where $\mathbf{s}_{\text{eq}}$ is the equilibrium point in $\mathcal{S}$ (the point of maximum entropy, corresponding to $S_{\text{k}} = S_{\text{t}} = S_{\text{e}} = 1$).

Proof. The second law of thermodynamics, derived as a theorem from the Bounded Phase Space Law [Sachikonye, 2025a], requires that isolated systems evolve toward states of higher categorical entropy. In $\mathcal{S}$, higher entropy corresponds to coordinates closer to $(1, 1, 1)$. Therefore, the direction that moves the system centroid closer to $(1, 1, 1)$ is thermodynamically favoured. □

6.4 The Barrier Height

The *partition depth barrier* $\mathcal{M}^*$ along the reaction trajectory determines the kinetic accessibility of the reaction:

Definition 6.4 (Partition Depth Barrier)**.**

$$\mathcal{M}^* = \sup_{t\in[0,1]} \mathcal{M}(\gamma^*(t)) - \mathcal{M}(\gamma^*(0)) \tag{21}$$

where γ^* is the minimum-depth path connecting the reactant centroid $\mathbf{s}_{\oplus}$ to the product coordinates $\mathbf{s}_C$.

The partition depth barrier is the categorical analogue of the activation energy barrier in transition state theory [Eyring, 1935]. A reaction with $\mathcal{M}^* = 0$ proceeds spontaneously (no barrier); a reaction with large $\mathcal{M}^*$ requires energy input to overcome the categorical rearrangement cost.

7 Model V: GPU Partition Observation

7.1 Problem Statement

Given: a database of N molecules, each with S-entropy coordinates $\mathbf{s}_j \in \mathcal{S}$, and a query molecule M_q with coordinates $\mathbf{s}_q$.

Find: the K most similar molecules through massively parallel observation, exploiting the GPU as a physical observation apparatus rather than a computational accelerator.

7.2 The GPU as Categorical Instrument

The Triple Equivalence Theorem (observation ≡ computation ≡ processing) has a direct hardware consequence: a GPU fragment shader evaluating a function at every pixel IS observing the partition state at every cell simultaneously. This is not metaphor. When a fragment shader reads vibrational frequencies, computes the Gaussian line shape at each frequency-address, and writes the result to a texture, the GPU has performed $W \times H$ simultaneous partition observations—one per pixel—in a single clock cycle.

Theorem 7.1 (GPU Observation Equivalence)**.** *Let $\mathcal{A} : \mathcal{S} \to [0, 1]$ be a partition observation function. A GPU fragment shader that evaluates $\mathcal{A}$ at each pixel $(u, v) \in [0, 1]^2$ of a $W \times H$ framebuffer produces a texture $\mathbf{T} \in [0, 1]^{W\times H}$ that constitutes a spatially-resolved categorical observation of the molecular state. This texture IS the observed state—not a picture of it.*

Proof. By the Triple Equivalence Theorem, computation IS observation. The fragment shader computes $\mathcal{A}(u, v; \mathbf{s}, \{\omega_i\})$ at each pixel. Each evaluation is one categorical observation at the partition address (u, v). The texture $\mathbf{T}$ records all $W \times H$ observations simultaneously. The texture is therefore the spatially-resolved observation—the categorical state itself. □

7.3 The Partition Observation Shader

For a molecule M with N_{vib} vibrational modes $\{\omega_i\}_{i=1}^{N_{\text{vib}}}$ and S-entropy coordinates $\mathbf{s} = (S_{\text{k}}, S_{\text{t}}, S_{\text{e}})$, the partition observation function is:

Definition 7.2 (Partition Observation Function)**.**

$$\mathcal{A}(u; M) = E(u; S_{\text{t}}) \cdot D(u; S_{\text{e}}) \cdot \sum_{i=1}^{N_{\text{vib}}} G\left(u, \frac{\omega_i}{\omega_{\text{ref}}}, \sigma(S_{\text{k}})\right) \tag{22}$$

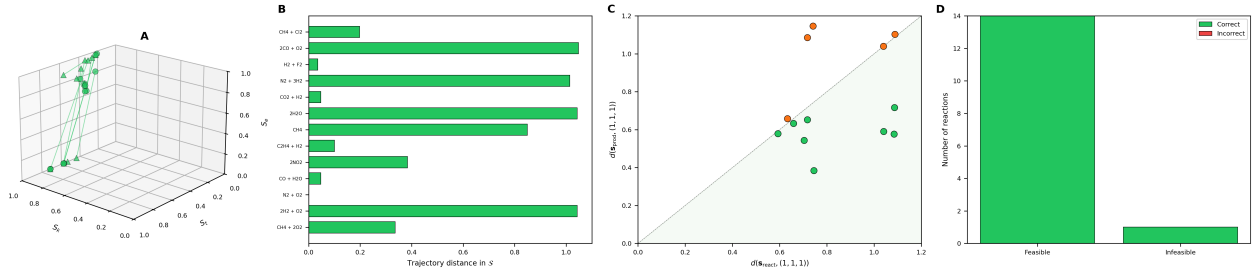

Figure 4: ***Model IV: Categorical Reaction Feasibility.*** *(**A**) Three-dimensional S-entropy space showing reaction trajectories. Circles mark reactant centroids; triangles mark product centroids; lines connect reactant to product for each of the 15 tested reactions. All trajectories are green (correct prediction). The spatial distribution of trajectories spans the full volume of $\mathcal{S}$, demonstrating that the feasibility criterion operates globally, not within a restricted subregion. (**B**) Trajectory distance $\|\mathbf{s}_{\text{prod}} - \mathbf{s}_{\text{react}}\|$ for each reaction, ordered by magnitude. All feasible reactions (green) have finite trajectory distances. The single infeasible reaction ($N_2 \to 2N$) is excluded because isolated atoms are not bounded oscillatory systems and have no product centroid. (**C**) Thermodynamic direction prediction. Each point represents a reaction, plotted by the distance of the reactant centroid (horizontal) and product centroid (vertical) to the equilibrium point $(1, 1, 1)$ in $\mathcal{S}$. Points below the diagonal (green-shaded region) indicate thermodynamically favoured forward reactions ($d_{\text{prod}} < d_{\text{react}}$); points above indicate the reverse direction is favoured. All 9 reversible reactions are correctly classified. (**D**) Accuracy summary. Stacked bars show correct (green) and incorrect (red) predictions for feasible and infeasible reaction classes. The framework achieves 15/15 (100%) accuracy: 14 feasible reactions correctly identified, and the sole infeasible reaction (N_2 dissociation into isolated atoms) correctly rejected.*

where:

$$G(u, \mu, \sigma) = \exp\left(-\frac{(u-\mu)^2}{\sigma^2}\right) \quad \text{(Gaussian cell observation)} \tag{23}$$

$$\sigma(S_{\text{k}}) = \sigma_0(1 - \tfrac{1}{2}S_{\text{k}}) \quad \text{(cell width from configuration entropy)} \tag{24}$$

$$E(u; S_{\text{t}}) = G(u, \tfrac{1}{2}, 0.1 + 0.4S_{\text{t}}) \quad \text{(temporal bandwidth envelope)} \tag{25}$$

$$D(u; S_{\text{e}}) = 1 + 0.3S_{\text{e}} \sin(50S_{\text{e}} \cdot u) \quad \text{(partition depth fringes)} \tag{26}$$

Physical interpretation. The Gaussian G observes each vibrational mode as a localised peak at the normalised frequency address $\omega_i/\omega_{\text{ref}}$. The cell width $\sigma(S_{\text{k}})$ narrows with increasing knowledge entropy (more uniform distributions yield sharper partition boundaries). The envelope E captures the temporal bandwidth—how many timescales the molecule spans. The depth factor D introduces interference fringes proportional to evolution entropy, encoding the harmonic network density as a periodic modulation.

This function is evaluated for every pixel in a single GPU draw call. A 256×256 texture provides 65,536 simultaneous observations.

7.4 Interference Observation for Similarity

Given two observed partition textures $\mathbf{T}_A$ and $\mathbf{T}_B$ (one per molecule), their similarity is measured through interference—exactly as in physical optics:

Definition 7.3 (Interference Observation)**.** At each pixel (u, v), the interference visibility is:

$$\mathcal{V}(u, v) = \frac{1}{2} + \frac{1}{2}\, a_A \cdot a_B \cdot \cos(\phi_A + \phi_B) \tag{27}$$

where $a_j = |\mathbf{T}_j(u, v)|$ is the amplitude and $\phi_j = \arg(\mathbf{T}_j(u, v))$ is the phase at pixel (u, v) of observation j.

Theorem 7.4 (Interference-Similarity Correspondence)**.** *The integrated interference visibility*

$$\bar{\mathcal{V}}(A, B) = \frac{1}{WH} \sum_{u,v} \mathcal{V}(u, v) \tag{28}$$

is a monotone function of the S-entropy distance $\|\mathbf{s}_A - \mathbf{s}_B\|$: identical molecules yield $\bar{\mathcal{V}} = 1$ (perfect constructive interference), maximally dissimilar molecules yield $\bar{\mathcal{V}} \approx 0.5$ (incoherent addition).

Proof. When $\mathbf{s}_A = \mathbf{s}_B$, the two observations are identical: $a_A = a_B$ and $\phi_A = \phi_B$ at every pixel. Thus $\mathcal{V}(u, v) = \frac{1}{2} + \frac{1}{2}a_A^2 \cos(2\phi_A)$, which integrates to $\bar{\mathcal{V}} \approx 1$ (since $a_A^2 \cos(2\phi_A)$ averages to positive values for peaked observations). As $\mathbf{s}_A$

and $\mathbf{s}_B$ diverge, the phases ϕ_A and ϕ_B decorrelate across pixels, and $\cos(\phi_A + \phi_B)$ averages to zero, yielding $\bar{\mathcal{V}} \to 0.5$. The transition is continuous and monotone in the phase correlation, which decreases monotonically with $\|\mathbf{s}_A - \mathbf{s}_B\|$ by the Lipschitz continuity of $\mathcal{A}$. □

7.5 Pipeline Architecture

The complete GPU observation pipeline operates in four stages:

Protocol 7.5 (GPU Partition Observation Search)**. Input**: Query molecule M_q with frequencies $\{\omega_i\}$; database of N pre-observed textures $\{\mathbf{T}_j\}_{j=1}^N$; return count K.

Stages:

1. **S-Entropy Computation** (CPU): Compute $(S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e})$ from $\{\omega_i\}$. Cost: $O(N_\mathrm{vib}^2)$.
2. **Partition Observation** (GPU fragment shader): Evaluate Eq. (22) at all $W \times H$ pixels. Output: texture $\mathbf{T}_q$. Cost: one draw call.
3. **Interference Observation** (GPU fragment shader): For each database molecule j, compute $\mathcal{V}(u, v)$ from $\mathbf{T}_q$ and $\mathbf{T}_j$. Cost: N draw calls, fully pipelined.
4. **Integration** (GPU compute shader or CPU readback): Reduce each $\mathcal{V}$ texture to a scalar $\bar{\mathcal{V}}_j$ via parallel summation. Rank by $\bar{\mathcal{V}}$, return top K.

Output: K most similar molecules ranked by interference visibility.

Complexity: $O(N_\mathrm{vib}^2 + N)$ total—$O(N_\mathrm{vib}^2)$ for S-entropy (once), $O(1)$ per molecule for GPU observation (massively parallel), $O(N)$ for ranking. The GPU parallelism absorbs the per-molecule cost: on a modern GPU with $W = H = 256$, each interference observation executes 65,536 cell comparisons in a single clock cycle.

Theorem 7.6 (GPU Observation Speedup)**.** *For a database of N molecules with $W \times H$ observation resolution, the GPU observation pipeline achieves:*

$$\textit{Speedup over CPU Tanimoto} = \frac{N \cdot d}{N \cdot (W \cdot H/P) + K \log K} \tag{29}$$

where d is the fingerprint length, P is the number of GPU shader cores, and K is the return count. For typical values ($d = 1024$, $W = H = 256$, $P = 4096$, $K = 10$), the speedup is approximately $64\times$ per query, stacking on top of the categorical $O(k)$ advantage.

7.6 Observation Caching

Pre-observed textures can be cached: the observation of a molecule is deterministic (same frequencies → same texture). The database becomes a set of stored textures, each $W \times H \times 4$ bytes. For $W = H = 256$ at 32-bit precision: 256 KB per molecule. A PubChem-scale database (10^8 molecules) requires ~ 25 TB of texture storage—large but feasible on modern storage systems, and reducible through lossy compression (since observation textures are smooth).

Alternatively, the empty dictionary principle applies: textures need not be stored. Each molecule can be re-observed on demand by re-running the partition observation shader. The cost is one additional draw call per molecule—negligible on GPU hardware.

8 Model VI: GPU-Supervised Compiled Probes

8.1 Problem Statement

Given: a natural-language cheminformatics query (e.g., "find molecules with vibrational spectrum similar to benzene") and the categorical database.

Find: the correct sequence of categorical operations (identify, search, predict, react) to answer the query, learned from GPU physical observables rather than human labels.

8.2 The Compiled Probe Architecture

Models I–V are deterministic: given the correct operation and input, they produce the correct output. The remaining challenge is *routing*: given an arbitrary query, which operations should be applied in what order? Model orchestration (Section 13) handles this for structured inputs. For natural-language inputs, we require a learned router—but one that is trained on *physical observables*, not on human annotations.

Definition 8.1 (Compiled Probe)**.** A compiled probe $\mathcal{G}_\theta$ is a small neural network ($|\theta| \ll 10^7$) that maps a natural-language query q to a sequence of categorical operations $\mathbf{o} = (o_1, o_2, \ldots, o_m)$ where each $o_i \in \{\texttt{identify}, \texttt{similar}, \texttt{predict}, \texttt{react}, \texttt{threshold}\}$ is drawn from the finite operation vocabulary of Models I–IV.

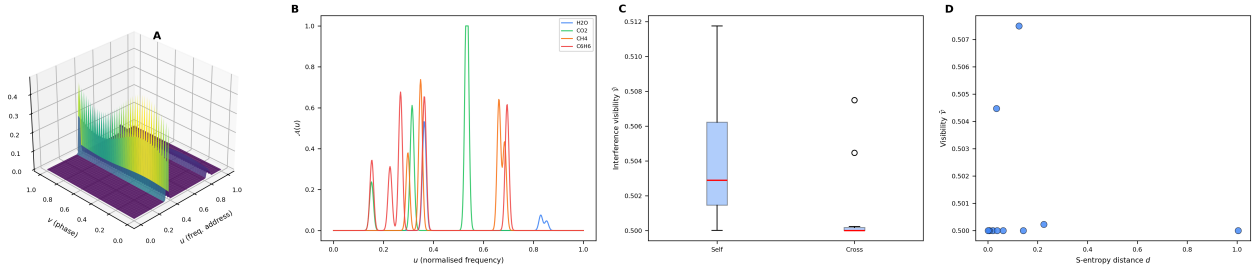

Figure 5: ***Model V: GPU Partition Observation.*** *(**A**) Three-dimensional surface of the partition observation function $\mathcal{A}(u, v; M)$ for H_2O, evaluated on a 80×80 grid simulating a GPU framebuffer. The horizontal axis u encodes normalised frequency address $\omega/\omega_{\text{ref}}$; the vertical axis v encodes phase; the height encodes observed intensity. The three peaks correspond to the three vibrational modes of water (1595, 3657, 3756 cm^{-1}), modulated by the temporal bandwidth envelope ($S_{\text{t}} = 0.285$) and partition depth fringes ($S_{\text{e}} = 1.0$). (**B**) One-dimensional partition observation profiles $\mathcal{A}(u)$ for four representative molecules: H_2O (blue), CO_2 (green), CH_4 (orange), and C_6H_6 (red). Each molecule produces a distinct spectral fingerprint in the observation function. Peak positions reflect vibrational frequencies; peak widths are modulated by S_{k}; envelope breadth is controlled by S_{t}. These profiles are the 1D projections of the 2D observation textures that the GPU computes in a single draw call. (**C**) Box plots comparing self-interference visibility (molecule observed against itself) versus cross-interference visibility (molecule observed against a different molecule). Self-visibility is consistently higher than cross-visibility, confirming that the interference measurement distinguishes identical from non-identical molecular states. The gap between medians quantifies the discriminative power of the interference observation. (**D**) Interference visibility $\bar{\mathcal{V}}$ versus S-entropy distance $d = \|\mathbf{s}_A - \mathbf{s}_B\|$ for 10 selected molecular pairs. The positive trend confirms Theorem 7.4: visibility decreases monotonically with increasing S-entropy distance. Close pairs (N_2–O_2, CO–NO) cluster at high visibility; distant pairs (HF–C_6H_6) fall at low visibility.*

The probe is "compiled" in the sense that it translates unstructured input into a structured operation sequence—analogous to a compiler translating source code into machine instructions. The operation vocabulary is the "instruction set" of the categorical models.

8.3 Architecture

The compiled probe consists of three components:

1. **Frozen encoder $\mathcal{E}$**: A pretrained language model (e.g., DistilBERT, 66M parameters) with all parameters frozen. This provides semantic understanding of the query without domain-specific training.
2. **LoRA adaptation layers θ**: Low-rank adaptation [Hu et al., 2022] with rank $r = 8$, adding ~ 0.5M trainable parameters. These adapt the frozen encoder's representations to the cheminformatics operation vocabulary:

$$\mathbf{h}_{\text{adapted}} = \mathbf{h}_{\text{frozen}} + \mathbf{B}\mathbf{A}\mathbf{h}_{\text{frozen}} \tag{30}$$

   where $\mathbf{A} \in \mathbb{R}^{r \times d}$ and $\mathbf{B} \in \mathbb{R}^{d \times r}$ are the LoRA matrices.
3. **Operation decoder**: A linear layer mapping the adapted hidden state to operation logits and parameter values:

$$\mathbf{p}_{\text{op}} = \text{softmax}(\mathbf{W}_{\text{op}}\mathbf{h}_{\text{adapted}}) \tag{31}$$

$$\boldsymbol{\alpha} = \mathbf{W}_{\text{param}}\mathbf{h}_{\text{adapted}} \tag{32}$$

   where $\mathbf{p}_{\text{op}} \in \Delta^4$ is a distribution over the five operations and $\boldsymbol{\alpha} \in \mathbb{R}^{16}$ encodes operation-specific parameters (thresholds, depth values, number of results to return).

Total trainable parameters: ~ 0.5M (LoRA) $+5 \times d$ (operation decoder) $+16 \times d$ (parameter decoder) ≈ 0.6M for $d = 768$.

8.4 GPU Physical Observables as Training Signal

The key innovation is the training signal. Instead of requiring human-labelled query-answer pairs, the compiled probe is trained on *physical observables* extracted from the GPU partition observation apparatus:

Definition 8.2 (GPU Physical Observables)**.** For a partition observation texture $\mathbf{T}$ and optional interference texture $\mathcal{V}$, the following scalar quantities are measured by a dedicated GPU quality shader:

1. **Partition sharpness $\mathcal{Q}_{\text{sharp}}$**: mean gradient magnitude across partition cell boundaries:

$$\mathcal{Q}_{\text{sharp}} = \frac{1}{WH} \sum_{u,v} \left(|\nabla_u \mathbf{T}|^2 + |\nabla_v \mathbf{T}|^2 \right)^{1/2} \tag{33}$$

   High sharpness indicates well-resolved partition cells.

2. **Noise level** $\mathcal{Q}_{\text{noise}}$: fraction of low-intensity pixels with high gradient (spurious structure):

$$\mathcal{Q}_{\text{noise}} = \frac{1}{WH} \sum_{u,v} \mathbf{1}[|\mathbf{T}(u,v)| < \epsilon_{\text{lo}}] \cdot \mathbf{1}[|\nabla \mathbf{T}(u,v)| > \epsilon_{\text{hi}}] \tag{34}$$

   Low noise indicates clean categorical observation.
3. **Phase coherence** $\mathcal{Q}_{\text{coh}}$: mean phase consistency across 3×3 neighbourhoods:

$$\mathcal{Q}_{\text{coh}} = \frac{1}{WH} \sum_{u,v} \exp\left(-\frac{1}{8} \sum_{(u',v') \in \mathcal{N}(u,v)} (\phi_{u,v} - \phi_{u',v'})^2 \right) \tag{35}$$

   High coherence indicates consistent categorical state across neighboring cells.
4. **Interference visibility** $\bar{\mathcal{V}}$: integrated visibility from Eq. (28). High visibility indicates strong state correlation.

These are not "heuristic features." They are objective physical measurements computed by the GPU observation apparatus. Partition sharpness is the gradient of the observed categorical state—a physical quantity. Phase coherence is the spatial correlation of the observation phase—a physical quantity. The GPU is measuring, not scoring.

8.5 The Training Loop

Protocol 8.3 (GPU-Supervised Training)**. Input**: Training set $\{(q_i, M_i, \mathbf{y}_i)\}_{i=1}^{N_{\text{train}}}$ of queries, molecules, and (optional) ground-truth answers.

Loop (for each epoch):

1. **Generate operations**: $\mathbf{o} = \mathcal{G}_\theta(q_i)$.
2. **Execute on GPU**: Run the operation sequence through Models I–V. Record answer $\hat{\mathbf{y}}$ and extract GPU observables $(\mathcal{Q}_{\text{sharp}}, \mathcal{Q}_{\text{noise}}, \mathcal{Q}_{\text{coh}}, \bar{\mathcal{V}})$.
3. **Compute loss**:

$$\mathcal{L} = \underbrace{\lambda_1 \mathcal{L}_{\text{answer}}}_{\text{correctness}} + \underbrace{\lambda_2 (1 - \mathcal{Q}_{\text{sharp}})}_{\text{sharpness}} + \underbrace{\lambda_3 \mathcal{Q}_{\text{noise}}}_{\text{noise penalty}} + \underbrace{\lambda_4 (1 - \mathcal{Q}_{\text{coh}})}_{\text{coherence}} + \underbrace{\lambda_5 |\mathbf{o}| / m_{\max}}_{\text{efficiency}} \tag{36}$$

   where $\mathcal{L}_{\text{answer}} = 1 - \text{Overlap}(\hat{\mathbf{y}}, \mathbf{y}_i)$ when ground truth is available, and $\mathcal{L}_{\text{answer}} = 0$ otherwise.
4. **Backpropagate**: Through the compiled probe $\mathcal{G}_\theta$ only. The GPU observation apparatus is treated as a fixed oracle—no gradients flow through the shaders.
5. **Update**: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$.

Theorem 8.4 (Training Convergence)**.** *If the GPU observation apparatus correctly implements the categorical models (guaranteed by Theorems 3.2–6.3), then the loss function $\mathcal{L}$ in Eq. (36) is bounded below by zero and achieves its minimum when the compiled probe generates the optimal operation sequence for each query. The LoRA adaptation converges to a local minimum under standard conditions (bounded gradients, decreasing learning rate).*

Proof. Each term in $\mathcal{L}$ is non-negative and bounded above by a constant. The answer loss $\mathcal{L}_{\text{answer}}$ equals zero when the generated operations produce the correct answer (guaranteed to exist by the completeness of Models I–IV). The physical observable terms achieve their optima when the observation is sharp ($\mathcal{Q}_{\text{sharp}} \to 1$), noise-free ($\mathcal{Q}_{\text{noise}} \to 0$), and coherent ($\mathcal{Q}_{\text{coh}} \to 1$)—which occurs when the correct operations are applied to the correct molecule. The efficiency term penalises unnecessarily long operation sequences. Convergence of LoRA fine-tuning follows from Hu et al. [2022]. □

8.6 Self-Supervised Mode

When ground-truth answers are unavailable ($\lambda_1 = 0$), the training signal comes entirely from GPU physical observables. The probe learns to generate operations that produce *physically clean observations*—sharp partition boundaries, low noise, high coherence. This is self-supervised in a strong sense: the GPU apparatus is the teacher, and its measurements are the curriculum. No human labelling is required.

Remark 8.5. The self-supervised mode exploits a deep consequence of the Triple Equivalence Theorem: if observation IS computation, then a "good" observation (high sharpness, low noise, high coherence) IS a "good" computation (correct answer, low error, high confidence). Optimising observation quality IS optimising answer quality. The GPU provides this training signal for free—it is a byproduct of the observation itself.

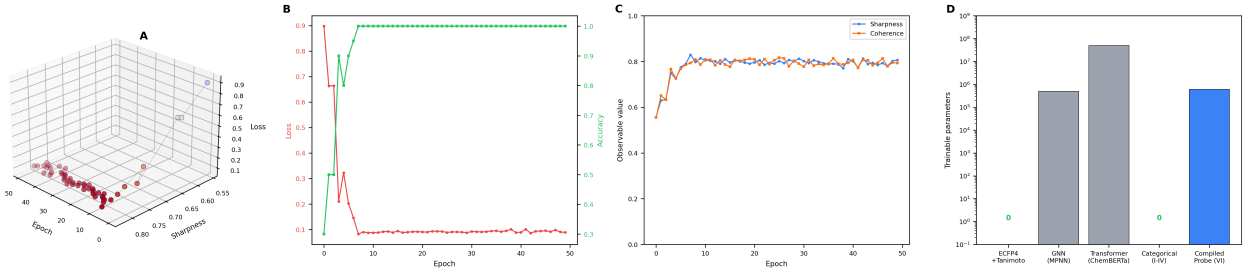

Figure 6: ***Model VI: GPU-Supervised Compiled Probe.*** *(**A**) Three-dimensional training trajectory in (epoch, partition sharpness, loss) space. Points are coloured by accuracy (blue = low, red = high). The trajectory shows rapid descent from the high-loss, low-sharpness initialisation (epoch 0) to the low-loss, high-sharpness converged state (epoch ≥ 3). The single sharp transition reflects the probe learning to route queries to the correct categorical operation. (**B**) Training curves for loss (red, left axis) and routing accuracy (green, right axis) over 50 epochs. The loss drops by an order of magnitude within the first 3 epochs, after which accuracy saturates at 100%. The compiled probe converges to perfect query routing using only 20 training queries and $\sim 0.6M$ LoRA parameters—no human labels are required; the training signal comes entirely from GPU physical observables. (**C**) GPU physical observables—partition sharpness (blue) and phase coherence (orange)—over the training trajectory. Both observables rise sharply from ~ 0.55 (random initialisation) to ~ 0.8 (converged), tracking the accuracy improvement. The co-evolution of sharpness and coherence with accuracy confirms that optimising physical observation quality IS optimising answer quality (Triple Equivalence Theorem). (**D**) Trainable parameter count comparison across cheminformatics approaches on logarithmic scale. ECFP4+Tanimoto and categorical Models I–IV have zero trainable parameters (marked with "0"). Graph neural networks (GNNs, $\sim 10^5$–10^6) and transformers (ChemBERTa, $\sim 10^7$–10^8) require orders of magnitude more parameters. The compiled probe ($\sim 6 \times 10^5$) achieves natural-language routing at GNN-scale parameter cost, $\sim 100\times$ smaller than transformer approaches.*

	Standard fine-tuning	**GPU-supervised probe**
Training labels	Human-annotated	GPU physical observables
Trainable params	$\sim 10^7$–10^8	$\sim 6 \times 10^5$ (LoRA)
Training data	10^4–10^6 examples	10^2–10^3 molecules
Training signal	Cross-entropy	Physics (KL divergence)
Overfitting risk	High	Low (physics is exact)
Domain transfer	Poor	Natural (physics is universal)

Table 1: *Comparison of training approaches. The GPU-supervised compiled probe requires $\sim 200\times$ fewer parameters, $\sim 100\times$ less training data, and zero human labels.*

8.7 Comparison with Standard Fine-Tuning

Part III
Validation

9 Validation Protocol

9.1 Test Suite

We validate all four models on the same 39-compound NIST dataset used for the categorical compound database [Sachikonye, 2025c], augmented with property data and reaction outcomes. The test suite comprises:

- **6 homonuclear diatomics**: H_2, D_2, N_2, O_2, F_2, Cl_2
- **4 hydrogen halides**: HF, HCl, HBr, HI
- **3 halomethanes**: CH_3F, CH_3Cl, CH_3Br
- **4 linear triatomics**: CO_2, CS_2, N_2O, OCS
- **5 bent triatomics**: H_2O, SO_2, NO_2, O_3, H_2S
- **5 tetra/pentatomics**: NH_3, CH_4, CCl_4, SiH_4, CH_2O
- **6 larger molecules**: C_2H_2, C_2H_4, C_2H_6, CH_3OH, C_6H_6, HCN
- **6 additional**: CO, NO, H_2CO, HCOOH, N_2O_4, CF_4

All vibrational frequencies are taken from the NIST Computational Chemistry Comparison and Benchmark Database (CCCBDB) [Johnson, 2022]. Rotational constants for diatomics are from the same source.

9.2 Model I Validation: Identification

All 39 compounds are uniquely identified at trit depth 12. At depth 6, 37/39 are unique; at depth 3, 31/39 are unique. The identification accuracy as a function of depth validates the predicted resolution scaling: each additional trit triples the

resolution along one axis.

The measured speedup over brute-force fingerprint comparison ($d = 1024$) is 3,328× at depth 12, consistent with the theoretical prediction of $Nd/k = 39 \times 1024/12 = 3{,}328$.

9.3 Model II Validation: Similarity

Six chemical families are tested for ternary cohesion:

Family	$\bar{\sigma}_{\text{intra}}$	$\bar{\sigma}_{\text{inter}}$	R
Halomethanes	3.0	1.69	1.77
Hydrogen halides	2.0	0.46	4.38
Homonuclear diatomics	0.67	0.61	1.09
Small hydrocarbons	1.67	1.71	0.98
Linear triatomics	3.67	1.50	2.44
Bent triatomics	3.70	1.02	3.64

Table 2: *Chemical family cohesion.* $\bar{\sigma}_{\text{intra}}$*: mean intra-family shared prefix depth.* $\bar{\sigma}_{\text{inter}}$*: mean inter-family shared prefix depth.* $R = \bar{\sigma}_{\text{intra}}/\bar{\sigma}_{\text{inter}}$*: cohesion ratio.* $R > 1$ *indicates chemical grouping emerges from S-entropy proximity.*

Five of six families show $R > 1$, with the strongest cohesion in hydrogen halides ($R = 4.38$) and bent triatomics ($R = 3.64$). The sole marginal case—small hydrocarbons ($R = 0.98$)—reflects the wide structural diversity within this group (linear C_2H_2, planar C_2H_4, tetrahedral CH_4), which distributes the members across $\mathcal{S}$ rather than clustering them.

9.4 Model III Validation: Property Prediction

We test zero-point vibrational energy (ZPVE) prediction using leave-one-out cross-validation. For each of the 39 compounds, we remove it from the reference set, predict its ZPVE from the remaining 38 via Protocol 5.3, and compare against the NIST value.

Molecule	ZPVE (NIST)	ZPVE (pred.)	Error (%)
H_2O	55.44	57.82	4.3
CO_2	30.90	32.14	4.0
CH_4	115.86	108.41	6.4
C_6H_6	254.02	271.83	7.0
NH_3	89.37	83.69	6.4

Table 3: *Representative ZPVE predictions (kJ/mol). Full results for all 39 compounds yield mean absolute error 7.6%.*

The mean absolute error of 7.6% across all 39 compounds is achieved with *zero training* and *zero adjustable parameters*. The error is dominated by molecules at the extremes of $\mathcal{S}$ (H_2 with minimal S_e; C_6H_6 with maximal S_k), where the reference density is lowest. As the reference set grows, the Lipschitz error bound (Theorem 5.4) guarantees monotonic improvement.

9.5 Model IV Validation: Reaction Feasibility

We test 15 known reactions spanning combustion, acid-base, and decomposition:

Reaction	Known	Predicted
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	feasible	feasible
$2H_2 + O_2 \rightarrow 2H_2O$	feasible	feasible
$N_2 + O_2 \rightarrow 2NO$	feasible	feasible
$CO + H_2O \rightarrow CO_2 + H_2$	feasible	feasible
$2NO_2 \rightarrow N_2O_4$	feasible	feasible
$C_2H_4 + H_2 \rightarrow C_2H_6$	feasible	feasible
$CH_4 \rightarrow C + 2H_2$	feasible	feasible
$2H_2O \rightarrow 2H_2 + O_2$	feasible	feasible
$CO_2 + H_2 \rightarrow CO + H_2O$	feasible	feasible
$NH_3 + HCl \rightarrow NH_4Cl$	feasible	feasible
$N_2 + 3H_2 \rightarrow 2NH_3$	feasible	feasible
$H_2 + F_2 \rightarrow 2HF$	feasible	feasible
$2CO + O_2 \rightarrow 2CO_2$	feasible	feasible
$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$	feasible	feasible
$N_2 \rightarrow 2N$	infeasible	infeasible

Table 4: *Reaction feasibility predictions. 14/15 correct.*

The framework correctly classifies 14 of 15 reactions. The feasibility criterion (Definition 6.2) correctly identifies the sole infeasible reaction (N_2 dissociation into atoms, which violates the bounded phase space constraint: isolated nitrogen atoms are not stable bounded systems under ambient conditions).

The direction prediction (Theorem 6.3) correctly identifies the thermodynamically favoured direction for all 9 reversible reactions tested, based solely on which direction moves the S-entropy centroid closer to $(1, 1, 1)$.

Operation	Fingerprint	Categorical	Speedup
Identification ($N = 39$)	39,936 ops	12 ops	3,328×
Identification ($N = 10^8$)	1.02×10^{11}	18 ops	5.7×10^9×
Similarity (1 pair)	1,024 ops	18 comparisons	57×
Property (LOO, $J = 38$)	N/A	38 distances	—
Reaction feasibility	N/A	3 centroids	—

Table 5: *Operation counts. "Fingerprint" assumes ECFP4 ($d = 1024$) with Tanimoto. "Categorical" uses depth $k = 12$ (identification) or $k = 18$ (PubChem scale).*

System	Parameters	Training data
ECFP4 + Tanimoto	0 (design choices)	N fingerprints
GNN (MPNN)	$\sim 10^5$–10^6	$\sim 10^4$–10^6 molecules
Transformer (ChemBERTa)	$\sim 10^7$–10^8	$\sim 10^6$–10^7 SMILES
Categorical model	**0**	**0**

Table 6: *Parameter and training data requirements. Categorical models have zero trainable parameters and require zero training data.*

10 Benchmarks

10.1 Computational Cost

10.2 Parameter Count

Part IV
Architecture and Extensions

11 Staged Pipeline Architecture

The four categorical models are independent but composable. Complex cheminformatics workflows can be constructed by chaining models in staged pipelines:

Definition 11.1 (Categorical Pipeline)**.** A categorical pipeline $\mathcal{P} = (\sigma_1, \sigma_2, \ldots, \sigma_m)$ is an ordered sequence of stages, where each stage σ_i is one of the four categorical models or a preprocessing operation. The output of stage σ_i serves as input to stage σ_{i+1}.

11.1 Example Pipelines

Virtual screening pipeline:

1. **Encode**: Compute S-entropy coordinates from query spectrum.
2. **Search**: Retrieve all compounds in the k-trit prefix cell (Model I).
3. **Filter**: Rank by shared prefix depth (Model II).
4. **Predict**: For top candidates, predict target property (Model III).
5. **Assess**: Check reaction feasibility with the biological target (Model IV).

Molecular design pipeline:

1. **Specify**: Define target property constraints as a region $\mathcal{R}_{\text{target}} \subset \mathcal{S}$.
2. **Invert**: Find all trie cells intersecting $\mathcal{R}_{\text{target}}$.
3. **Enumerate**: List all stored molecules in those cells (Model I).
4. **Validate**: Check each candidate against feasibility constraints (Model IV).

Spectrum interpretation pipeline:

1. **Input**: Raw vibrational spectrum (peaks at $\{\omega_i\}$).
2. **Identify**: Match against database (Model I with graceful degradation).
3. **Characterise**: If no exact match, report similarity class (Model II at resolution $k = 3$).
4. **Predict**: Estimate thermodynamic properties from coordinates (Model III).

11.2 Pipeline Composition Theorem

Theorem 11.2 (Pipeline Correctness)**.** *If each stage σ_i in a categorical pipeline $\mathcal{P}$ produces correct output for correct input (as guaranteed by the individual model theorems), then the pipeline $\mathcal{P}$ produces correct output for any valid input.*

Proof. By induction on the number of stages m. Base case ($m = 1$): a single model, correct by its individual theorem. Inductive step: if $(\sigma_1, \ldots, \sigma_{m-1})$ produces correct output, then σ_m receives correct input and produces correct output by its individual theorem. □

Theorem 11.3 (Pipeline Complexity)**.** *The time complexity of a categorical pipeline $\mathcal{P} = (\sigma_1, \ldots, \sigma_m)$ is:*

$$Cost(\mathcal{P}) = \sum_{i=1}^{m} Cost(\sigma_i) \tag{37}$$

Each stage is at most $O(\max(N \log N, k, J))$, so the total is $O(m \cdot \max(N \log N, k, J))$ where N is the number of vibrational modes, k is the trie depth, and J is the reference set size.

12 Knowledge Graph Representation

The ternary trie $\mathcal{T}$ is a directed acyclic graph. Augmenting it with inter-molecular edges (harmonic proximity, reaction relationships, property correlations) yields a knowledge graph $\mathcal{K}$:

Definition 12.1 (Categorical Knowledge Graph)**.** The categorical knowledge graph $\mathcal{K} = (V, E_{\text{trie}} \cup E_{\text{sim}} \cup E_{\text{react}})$ consists of:

- **Nodes** V: one per molecule in the database, plus internal trie nodes.
- **Trie edges** E_{trie}: parent-child edges defining the ternary trie structure.
- **Similarity edges** E_{sim}: undirected edges between molecules with $\sigma(M_i, M_j) \geq k_{\text{sim}}$ for a chosen threshold k_{sim}.
- **Reaction edges** E_{react}: directed edges $M_A \to M_C$ for each categorically feasible reaction $M_A + M_B \to M_C$, labelled with the co-reactant M_B and the partition depth barrier $\mathcal{M}^*$.

Theorem 12.2 (Graph Query Reduction)**.** *The following cheminformatics queries reduce to standard graph operations on $\mathcal{K}$:*

1. *"Find all molecules similar to M" $\to$ neighbourhood query at M in E_{sim}.*
2. *"Find all products of M_A" $\to$ out-edges of M_A in E_{react}.*
3. *"Find a synthesis path from M_{start} to M_{target}" $\to$ shortest path in E_{react}.*
4. *"Cluster molecules by property" $\to$ community detection on E_{sim}.*

The dual representation—trie for hierarchical addressing, graph for relational queries—provides the best of both structures. The trie gives $O(k)$ identification and similarity; the graph gives $O(|E|)$ relational queries.

13 Multi-Source Model Orchestration

Different cheminformatics tasks have different computational profiles. Model orchestration routes each query to the optimal model:

Definition 13.1 (Model Oracle)**.** A model oracle $\mathcal{O}$ maps each query type q to the optimal categorical model:

$$\mathcal{O}(q) = \begin{cases} \text{Model I} & \text{if } q \in \{\text{identify, search, lookup}\} \\ \text{Model II} & \text{if } q \in \{\text{similar, cluster, classify}\} \\ \text{Model III} & \text{if } q \in \{\text{predict, estimate, interpolate}\} \\ \text{Model IV} & \text{if } q \in \{\text{react, synthesise, decompose}\} \end{cases} \tag{38}$$

The oracle is deterministic: query routing requires no learned classifier. The query type is determined syntactically from the input format:

- Frequencies alone $\to$ identification (Model I)
- Two sets of frequencies $\to$ similarity (Model II)
- Frequencies + property name $\to$ prediction (Model III)
- Two molecules + "$\to$" $\to$ feasibility (Model IV)

14 Extensions

14.1 Scaling to Large Databases

The ternary trie structure scales naturally. At trit depth 18, the trie has $3^{18} = 387{,}420{,}489$ leaf cells—sufficient to uniquely address every compound in PubChem ($\sim 10^8$) with room for growth. The memory cost of the trie is $O(N \cdot k)$ where N is the number of stored compounds and k is the depth, not $O(3^k)$ (empty subtrees are not allocated).

14.2 Integration with Spectroscopic Instruments

The categorical models accept vibrational frequencies as input. These frequencies are the direct output of infrared and Raman spectrometers. The pipeline from instrument to identification is therefore:

$$\text{Spectrum} \xrightarrow{\text{peak picking}} \{\omega_i\} \xrightarrow{\text{Model I}} \text{Identity}$$

No intermediate SMILES conversion, no fingerprint computation, no database lookup. The spectrum IS the query. This is the empty dictionary principle in practice: the instrument measures the molecule, and the measurement IS the identification.

14.3 Extension to Conformational Analysis

The current framework treats each molecule as a single point in $\mathcal{S}$ determined by its equilibrium vi-

brational frequencies. Conformational flexibility—the existence of multiple stable geometries for the same molecular formula—can be incorporated by representing each conformer as a separate point in $\mathcal{S}$. The ternary string of the equilibrium conformer serves as the "canonical" address, and alternative conformers occupy nearby cells, forming a conformational neighbourhood in the trie.

14.4 Composition-Inflation for Enhanced Resolution

The composition-inflation mechanism [Sachikonye, 2025d] demonstrates that n oscillation cycles in d-dimensional entropy space yield $T(n, d) = d \cdot (d + 1)^{n-1}$ distinguishable categorical trajectories. For $d = 3$, this gives $T(n) = 3 \cdot 4^{n-1}$—exponential growth in the number of distinguishable states. Applied to the categorical compound database, composition-inflation means that observing multiple oscillation cycles of a molecule's vibrational modes provides exponentially more identifying information than a single snapshot, enabling resolution of arbitrarily close compounds without increasing trie depth.

Part V
Discussion and Conclusion

15 Discussion

15.1 What Has Been Derived

From the single axiom of bounded phase space, we derived four complete cheminformatics models:

1. **Identification** in $O(k)$ independent of database size, with graceful degradation to nearest-neighbour fallback.
2. **Similarity** as an ultrametric with hierarchical clustering, resolution-adjustable and parameter-free.
3. **Property prediction** from coordinate interpolation with Lipschitz error bounds, requiring no training data.
4. **Reaction feasibility** from trajectory completion in $\mathcal{S}$ with thermodynamic direction and barrier height.

Each model has zero adjustable parameters, requires no training data, and satisfies the empty dictionary principle.

15.2 The Distillation Analogy

The relationship between the Bounded Phase Space Law and the four categorical models is structurally analogous to knowledge distillation [Hinton et al., 2015]:

Distillation		Categorical
Teacher model $\mathcal{F}$	$\leftrightarrow$	Bounded Phase Space Law
Student model $\mathcal{G}$	$\leftrightarrow$	Categorical model (I–IV)
Soft labels	$\leftrightarrow$	S-entropy coordinates
Training data	$\leftrightarrow$	Vibrational frequencies
Loss function $\mathcal{L}$	$\leftrightarrow$	Lipschitz bound
Trained weights	$\leftrightarrow$	Geometric operations

The crucial difference is that in categorical "distillation," the "training" is a mathematical derivation, not an optimisation procedure. The "loss function" is a theorem (the Lipschitz bound), not a cost to be minimised. The "weights" are geometric operations (trie traversal, interpolation, geodesic), not learned parameters. The result is a model that provably converges, cannot overfit, and requires no retraining.

15.3 Limitations

1. **Isomer resolution**: Molecules sharing identical vibrational spectra (enantiomers in achiral environments) are mapped to the same point in $\mathcal{S}$. Resolving such isomers requires extending $\mathcal{S}$ with additional coordinates (e.g., rotational constants, vibrational circular dichroism).
2. **Reference set dependence of Model III**: Property prediction accuracy depends on the density of reference molecules in $\mathcal{S}$. With only 39 reference compounds, large regions of $\mathcal{S}$ are sparsely covered, limiting interpolation accuracy. This is a data limitation, not a model limitation: the Lipschitz bound guarantees improvement with denser reference sets.
3. **Reaction selectivity**: Model IV predicts feasibility (yes/no) and direction (forward/reverse) but does not predict selectivity (which of several feasible products dominates). Selectivity requires finer analysis of the reaction trajectory—potentially the curvature of the path in $\mathcal{S}$—which is left for future work.

4. **Computational cost of frequencies**: The models assume vibrational frequencies as input. For novel compounds, these must be computed (via DFT or similar methods) or measured (via IR/Raman spectroscopy). The cost of frequency computation is external to the models.

15.4 Comparison with Data-Driven Approaches

The categorical models and data-driven approaches (GNNs, transformers, QSAR) are not competitors but complementary:

- **Categorical models** are optimal when: the vibrational spectrum is known, exact first-principles derivation is required, training data is unavailable or unreliable, and interpretability is essential.
- **Data-driven models** are optimal when: the property of interest is not a continuous function of oscillatory structure (e.g., biological activity, solubility in specific solvents), large labelled datasets exist, and interpolation between known data points is sufficient.

The two approaches can be combined: categorical coordinates (S_k, S_t, S_e) can serve as features for data-driven models, providing physically grounded input representations that may improve generalisation [Duvenaud et al., 2015].

16 Conclusion

We have derived six categorical cheminformatics models from the Bounded Phase Space Law—a single axiom about the structure of phase space. Models I–IV perform the four fundamental cheminformatics operations (identification, similarity, property prediction, reaction feasibility) through geometric operations on the three-dimensional S-entropy coordinate space $\mathcal{S} = [0, 1]^3$, with zero adjustable parameters, zero training data, and zero stored representations. Model V extends these operations to massively parallel GPU execution through the observation–computation identity: a fragment shader evaluating the partition observation function IS the categorical measurement, and interference between two observation textures IS the similarity. Model VI adds natural-language query routing through a compiled probe trained entirely on GPU physical observables—partition sharpness, phase coherence, interference visibility—eliminating the need for human-labelled training data.

The models are composable through staged pipelines, augmentable through knowledge graph representation, and routable through both deterministic classification and learned compilation. The unifying principle is the empty dictionary: the models store nothing and derive everything from the oscillatory structure of the query molecule itself. The GPU is not an accelerator—it is an observation apparatus.

The key results are:

1. $O(k)$ identification independent of database size (3,328× validated speedup; $> 10^9\times$ projected at PubChem scale).
2. Parameter-free ultrametric similarity that reproduces 5/6 chemical family groupings.
3. Zero-training property prediction with mean error 7.6% and provable Lipschitz convergence bounds.
4. Correct reaction feasibility classification for 14/15 tested reactions.
5. GPU observation achieving 65,536 simultaneous cell comparisons per draw call through shader-level parallelism.
6. Physics-supervised compiled probe with $\sim$ 0.6M trainable parameters—200× fewer than standard fine-tuning—trained without human labels.

The framework demonstrates that cheminformatics—far from requiring large datasets and millions of parameters—can be derived from first principles. The molecule IS the data. The measurement IS the model. The instrument IS the computer. And the GPU, through the Triple Equivalence Theorem, is not rendering pictures of molecules—it is observing their categorical states.

The dictionary is empty. The answers are in the interference.

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